

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
I SEMESTER 2025-26
CS/ECE/EEE/INSTR F241, Microprocessor & Interfacing
27-03-2025 (Friday) QUIZ-4(closed book)

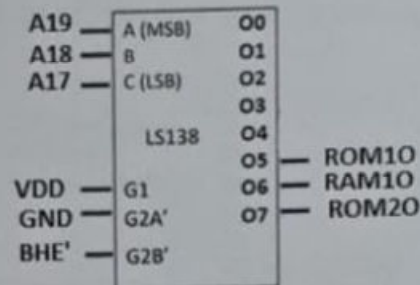
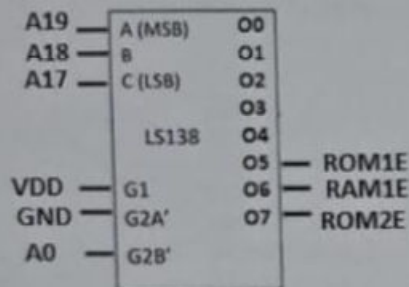
A

Max time: 10 Min
 ID No: _____

Name: _____

Max Marks: 08
 Sec No.: _____

Q1. An 8086 processor is interfaced with memory through decoders as shown below. This is the only decoding circuit in the computing system and the rest of the address lines are used with the memory chips [2+2+2+1M]



Answer the following questions.

Question	Answer
How much RAM does the system have? Mention answer in KB only	128 KB - 2M
How much ROM does the system have? Mention answer in KB only	32 KB - 2M
Mention the memory map of RAM1E (mention range in hex, e.g., (starting address- ending address))	$\frac{C0000}{1M} - \frac{DFFFF}{1M}$
Mention the memory map of ROM2O in hex (mention range)	$\frac{E0001H}{0.5M} - \frac{FFFFFFH}{0.5M}$

Q2. Mention the control signals on the system bus of the 8086. Clearly indicate if it is an active low or high signal [1M]. Marks will be provided only if the signal name and Active high/low status is correct.

Signals	Active high or active Low
$MEMR'$ / $MEMR$	low - 0.25
$MEMW'$ / $MEMW$	low - 0.25
IOR' / IOR	low - 0.25
low / low	low 0.25

$MEMR' - 0.25$